LESSON PLAN · PYTHON (GR 6–8) BEGINNER: PYTHON GAME CODERS

Lesson 5: Loops: Repeat Like a Robot

Module 2 | 60 minutes | for, range, while | Grades 6–10 · ages 11–15

Objective: Repeat code with for loops and range(), keep going with while loops, and avoid infinite loops.

Success criteria

- I can repeat code with a for loop and range().
- I can count up, count down and count in steps.
- I can use a while loop and make sure it stops.
- I can fix an infinite loop.

Key vocabulary

- **loop**: Code that repeats.
- **for loop**: Repeats once for each item, such as each number in a range.
- **range()**: Makes a list of numbers: range(1, 6) is 1 to 5.
- **while loop**: Repeats as long as a condition is True.
- **infinite loop**: A loop that never stops because its condition stays True.
- **body**: The indented lines inside a loop.

You will need

- Projector with Kitabash (lesson video, Python Lab)
- Devices: 1 per learner, with a keyboard
- Printed worksheet (optional)
- Project files: loop_practice.py

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game: Loop Dance**	Play Loop Dance (unplugged). Compare the clapping for loop with the jumping while loop.
5 min	**Hook and video**	Show the hook. Chat poll: “How many lines would 100 cheers take without a loop?” Play the video with captions.
10 min	**Live demo and build-along**	Run the live demo, then type the loop practice build-along together. Thumbs check after each mini-loop.
8 min	**Challenge, predict and sort**	Times-table printer, then the range predict and the for-or-while sort. Quick competition: first correct prediction of range(1, 10, 3).
15 min	**Project in breakout pairs**	Pairs build the rocket launch + table trainer. Partners test each other’s trainer with one wrong answer on purpose.
6 min	**Debug race**	Race to fix “the endless countdown”. Ask: why did the Lab stop it?

7 min	**Show-and-tell**	Three volunteers run their rocket launch; the class counts down out loud together. Share the best trainer score.
4 min	**Wrap-up and exit ticket**	Fist-to-five on the success criteria. Exit ticket: 10, 20, 30, 40, 50 loop.

Unplugged warm-up

Loop Dance (5 min game): the teacher writes “for i in range(4): clap, stamp” on the board and the class performs it. Then “while the teacher’s hand is up: jump”. Learners notice the for loop has a fixed count, but the while loop only stops when something changes.

Teaching the idea

Loops repeat code. A for loop with range() repeats a known number of times, and the loop variable counts for us. range(start, stop, step) stops before the stop number. A while loop repeats while a condition is True, so something inside the loop must change, or we get an infinite loop. The Python Lab stops runaway loops with an error message, which makes it a safe place to make this mistake.

Analogy: A for loop is like doing 10 skips with a skipping rope: you know the count. A while loop is like waiting at the bus stop: you keep waiting until the bus comes.

Live demo: 1) Type print("Hip hip hooray!") three times, then delete them and type for i in range(3): with one indented print. Run both versions and compare. 2) Change it to print(i) to show 0, 1, 2. 3) Type for n in range(1, 6): print(n, "x 4 =", n * 4) and run. 4) Ask the class to predict range(10, 0, -1), then run it. 5) Type count = 5, while count > 0:, print(count), count -= 1, then print("Blast off!") at the left edge. Run. 6) Delete count -= 1 and run: the Lab stops the endless 5s with an error. Put the line back.

Questions to ask

Ask	Listen for
How many times does range(5) repeat?	5 times (0 to 4).
What numbers does range(1, 4) give?	1, 2 and 3.
How do you count backwards from 10 to 1?	range(10, 0, -1).
Why did the countdown never stop?	count never changed, so count > 0 stayed True.
When would you use while instead of for?	When you do not know how many repeats you need.

Differentiation

Support
Give a card with three loop shapes to copy: for i in range(N), for i in range(A, B), while x > 0 with x -= 1.

Stretch
Print the full table as a grid using two loops, or make the countdown pause by printing a line of dots that gets shorter each time ("." * count).

Common misconceptions

Misconception	How to fix it
range(1, 10) includes 10	range stops before the stop number, so it ends at 9.
The loop variable starts at 1	range(5) starts at 0 unless you give a start.
A while loop changes its own variable	You must change it inside the loop yourself.
Lines after the loop repeat too	Only the indented lines repeat. Lines at the left edge run once, after the loop.

Assessment

Use the Python Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write a for loop that prints the numbers 10, 20, 30, 40, 50.

Homework
Use your table trainer for the table you find hardest. Then change it to ask questions 6 to 10, and record your score each day this week.

Answer key — Kitabash activities

1. Python Lab: times-table printer (perform · python lab)

Goals checked: Prints the 7 times table · Works for any table · Uses a for loop with range

One solution:

```
n = int(input("Which table? "))
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

2. Predict the output: range with a step (perform · code output)

3. Which loop fits best? (perform · categorize)

for loop (I know how many times): Print a cheer 10 times, Print the 9 times table up to 9 × 10, Go through the numbers 1 to 100
while loop (keep going until something happens): Keep asking for the password until it is right, Keep playing until lives reach 0, Keep asking until the user types quit

4. Project: rocket launch and table trainer (perform · python lab)

Goals checked: Counts down 10 to 1, then blast off · All right answers score 5 · Wrong answers score 0 · Uses while and for

Example program:

```
# Rocket launch + times-table trainer
# Made by: Kabir

# Part 1: the countdown
print("=== KITABASH SPACE CENTRE ===")
count = 10
while count > 0:
    print(count)
    count -= 1
print("Blast off! 🚀")
print()

# Part 2: the times-table trainer
table = int(input("Which times table do you want to practise? "))
score = 0
for i in range(1, 6):
    answer = int(input(f"{table} x {i} = "))
    if answer == table * i:
        print("Correct!")
        score += 1
    else:
        print(f"Oops, it is {table * i}")

print(f"You got {score} out of 5.")
if score == 5:
    print("Perfect! You are ready for orbit.")
```

5. Debug it: the endless countdown (probe · python lab)

Goals checked: Stops by itself · Prints 5 to 1 then Go!

One solution:

```
count = 5
while count > 0:
    print(count)
    count -= 1
print("Go!")
```

Answer key — worksheet

1. **Key words:** loop = Code that repeats. · for loop = Repeats once for each item, such as each number in a range. · range() = Makes a list of numbers: range(1, 6) is 1 to 5. · while loop = Repeats as long as a condition is True.
2. **Plan your program:** Goals: Prints the 7 times table; Works for any table; Uses a for loop with range. One solution: n = int(input("Which table? ")) for i in range(1, 11): print(n, "x", i, "=", n * i)

3. **Predict the output: range with a step:** A. 1, 4 and 7 on separate lines
4. **Which loop fits best?:** for loop (I know how many times): Print a cheer 10 times, Print the 9 times table up to 9 × 10, Go through the numbers 1 to 100 · while loop (keep going until something happens): Keep asking for the password until it is right, Keep playing until lives reach 0, Keep asking until the user types quit
5. **Plan your program:** Goals: Stops by itself; Prints 5 to 1 then Go!. One solution: count = 5 while count > 0: print(count) count -= 1 print("Go!")

Kitabash original curriculum.